

GUILLAIN-BARRÉ SYNDROME (GBS)

1. PATHOPHYSIOLOGY OF GUILLAIN-BARRÉ SYNDROME (GBS)

- **Guillain-Barré Syndrome (GBS)** is an **autoimmune disorder** (when the immune system mistakenly attacks the body).
- The immune system attacks the **myelin sheath** (protective covering of nerves) in the **peripheral nervous system** (nerves outside the brain and spinal cord).
- Causes **progressive muscle weakness and paralysis**.
- **Triggering events:**
 - **Recent infection** (e.g., respiratory or gastrointestinal infection, usually *Campylobacter jejuni* bacteria).
 - **Recent vaccination** (rare, but can follow flu vaccine).
 - **Surgery or trauma**.

2. SYMPTOMS OF GUILLAIN-BARRÉ SYNDROME

- **Ascending paralysis** (muscle weakness starting in the legs and moving upward).
- **Paresthesia** (tingling, numbness).
- **Loss of reflexes (areflexia)** – absent deep tendon reflexes).
- **Difficulty breathing** (if paralysis reaches the diaphragm).
- **Dysphagia** (trouble swallowing).
- **Autonomic dysfunction** (affects involuntary functions like heart rate and blood pressure):
 - **Tachycardia or bradycardia** (fast or slow heart rate).
 - **Fluctuating blood pressure**.
 - **Urinary retention** (unable to urinate).

3. DIAGNOSIS OF GUILLAIN-BARRÉ SYNDROME

- **Lumbar puncture (spinal tap):**
 - **↑ CSF protein** (increased cerebrospinal fluid protein without increased WBCs).
- **Electromyography (EMG) and nerve conduction studies:**
 - Show **slowed nerve signals**.

4. TREATMENT OF GUILLAIN-BARRÉ SYNDROME

ACUTE PHASE (HOSPITALIZATION REQUIRED)

- **Plasmapheresis** (removes harmful antibodies from the blood).
- **IV immunoglobulin (IVIG)** (helps block the immune attack).
- **Ventilator support** if respiratory muscles are weak.

SUPPORTIVE CARE

- **Monitor respiratory status** (priority due to risk of respiratory failure).
- **Prevent complications of immobility:**
 - **DVT prophylaxis** (prevention of blood clots with anticoagulants, compression stockings).
 - **Frequent repositioning** (to prevent pressure ulcers).
- **Physical therapy** (to regain strength).

5. NCLEX PRIORITIES FOR GUILLAIN-BARRÉ SYNDROME

AIRWAY & BREATHING (TOP PRIORITY!)

- ✓ **Monitor respiratory function** (ABGs, lung sounds, oxygen levels).
- ✓ **Prepare for intubation if breathing weakens.**
- ✓ **Assess ability to swallow** (prevent aspiration).

NEUROMUSCULAR & MOBILITY CARE

- ✓ **Turn every 2 hours** (prevent bedsores and pneumonia).
- ✓ **Perform passive range-of-motion exercises.**
- ✓ **Prevent foot drop with splints.**

AUTONOMIC DYSFUNCTION CARE

- ✓ **Monitor for abnormal heart rate or BP swings.**
- ✓ **Manage urinary retention** (may need catheterization).

COMPLICATIONS TO WATCH FOR

⚠ **Respiratory failure** (paralysis reaches the diaphragm):

- **Shallow breathing, low oxygen, inability to cough.**
 - ⚠ **Aspiration pneumonia** (from weak swallowing muscles):
- **Coughing while eating, drooling.**
 - ⚠ **Deep vein thrombosis (DVT) and pressure ulcers** (from immobility).